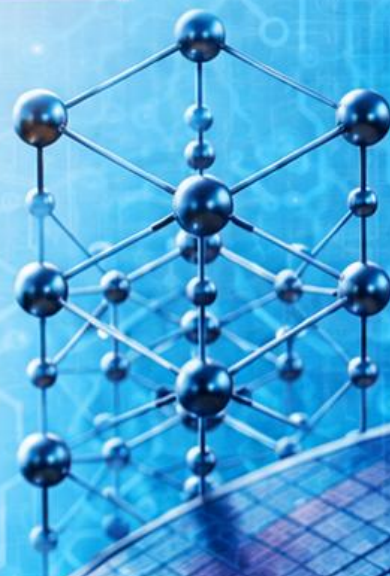


XII INTERNATIONAL CONFERENCE

Topical PROBLEMS — of — SEMICONDUCTOR PHYSICS



Prykarpattya, Drohobych,
UKRAINE, May 25–28, 2026



Ministry of Education and Science of Ukraine

Institute of Physics of NASU

V.E. Lashkaryov Institute of Semiconductor Physics NAS of Ukraine

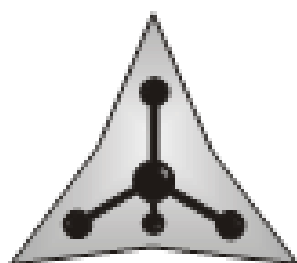
**Scientific Council “Semiconductor and Dielectric Physics” at Physics and
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XII-th International Conference

TOPICAL PROBLEMS OF SEMICONDUCTOR PHYSICS



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Actual problems and important achievements of modern semiconductors physics are presented in the Proceedings of the XII-th International Conference “Topical Problems of Semiconductors Physics”. The abstracts are grouped into 7 sections, according to the Conference Thematic: “A. New frontiers in semiconductors and their based structures for electronics, optoelectronics, spintronic and sensing”, “B. Semiconductor low-dimensional structures: advances in synthesis, characterization, theoretical modeling and applications”, “C. The semiconductors for LEDs, solar and related energy technologies and sensor materials”, “D. Synthesis, processing and characterization of multifunctional oxide materials”, “E. Advanced strategies for smart functional and multifunctional bionanomaterials and biointerfaces”, “F. Laser material processing: from fundamental interactions to innovative applications”, “G. Modern computational methods and their applications in materials science: Synergy of theory and experiment”. The Proceedings were prepared for publication by the Conference Program Committee and presented in the author’s edition.

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Physics-Informed Neural Network Prediction of Thermal Conductivity in Clustered Si–Ge Two-Phase Systems

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The study of thermal conductivity (TC) in two-phase systems is essential for tailoring thermal properties of advanced materials with complex microstructures. Si–Ge systems are particularly relevant as models of media with strong contrasts in atomic mass and elastic properties between phases.

In this work, TC was evaluated using a Physics-Informed Neural Network (PINN) based on the framework in [1]. The input was a 2D spatial distribution of TC components, while an encoder–decoder architecture predicted the

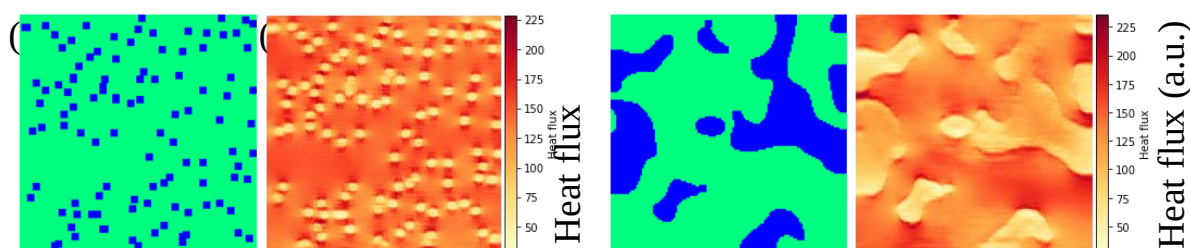


Fig. 1. Initial phase distributions (a, c) and predicted heat flux fields (b, d). (a, b) Ge fraction 0.1, clustering degree 0; (c, d) Ge fraction 0.3, clustering degree

temperature field and heat flux components. The loss function incorporated Fourier's law, governing equations, and boundary conditions.

TC of a two-phase Si–Ge system was investigated as a function of Ge fraction and clustering degree. Fig. 1 shows representative initial morphologies and corresponding heat flux distributions under a temperature gradient between the left and right boundaries. The resulting values are summarized in Fig. 2. The effective TC may fall below that of the constituent phases and exhibits a nonmonotonic dependence on clustering degree at intermediate fraction values.

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1. Xu J., Wei H., Bao H. Physics-Informed Neural Networks for Studying Heat Transfer in Porous Media. *Int. J. Heat Mass Transf.* 2023. V.217, 124671.

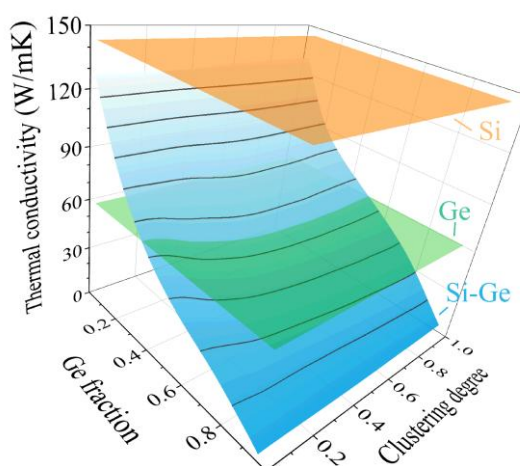


Fig. 2. Dependence of the TC of two-phase Si–Ge system on Ge fraction and clustering degree